

Integration by Parts and Recurrence Relations — Solutions

Warm-up

Question 1:

To evaluate $\int x \ln x \, dx$, we use integration by parts. Let $u = \ln x$ and $dv = x \, dx$.

Then, we have:

$$\begin{aligned} du &= \frac{1}{x} dx \\ dv = x \, dx &\implies v = \frac{1}{2} x^2 \end{aligned}$$

Applying the integration by parts formula $\int u \, dv = uv - \int v \, du$:

$$\begin{aligned} \int x \ln x \, dx &= \ln x \left(\frac{1}{2} x^2 \right) - \int \frac{1}{2} x^2 \left(\frac{1}{x} \right) dx \\ &= \frac{1}{2} x^2 \ln x - \int \frac{1}{2} x \, dx \\ &= \frac{1}{2} x^2 \ln x - \frac{1}{4} x^2 + C \end{aligned}$$

Question 2:

To evaluate $\int x \sin x \, dx$, we use integration by parts. Let $u = x$ and $dv = \sin x \, dx$.

Then, we have:

$$\begin{aligned} du &= dx \\ dv = \sin x \, dx &\implies v = -\cos x \end{aligned}$$

Applying the integration by parts formula $\int u \, dv = uv - \int v \, du$:

$$\begin{aligned} \int x \sin x \, dx &= x(-\cos x) - \int (-\cos x) dx \\ &= -x \cos x + \int \cos x \, dx \\ &= -x \cos x + \sin x + C \end{aligned}$$

Question 3:

To evaluate $\int \ln x \, dx$, we use integration by parts by treating the integrand as $\ln x \cdot 1$.

Let $u = \ln x$ and $dv = 1 \, dx$.

Then, we have:

$$\begin{aligned} du &= \frac{1}{x} dx \\ dv = 1 \, dx &\implies v = x \end{aligned}$$

Applying the integration by parts formula $\int u \, dv = uv - \int v \, du$:

$$\begin{aligned} \int \ln x \, dx &= x \ln x - \int x \left(\frac{1}{x} \right) dx \\ &= x \ln x - \int 1 \, dx \\ &= x \ln x - x + C \end{aligned}$$

Question 4:

To evaluate $\int_0^\pi e^x \sin x \, dx$, we use integration by parts twice.

Let $u = \sin x$ and $dv = e^x \, dx$.

$$\begin{aligned} du &= \cos x \, dx \\ dv = e^x \, dx &\implies v = e^x \end{aligned}$$

Applying integration by parts:

$$\int e^x \sin x \, dx = e^x \sin x - \int e^x \cos x \, dx$$

Now, we apply integration by parts again to $\int e^x \cos x \, dx$. Let $u = \cos x$ and $dv = e^x \, dx$.

$$\begin{aligned} du &= -\sin x \, dx \\ dv = e^x \, dx &\implies v = e^x \end{aligned}$$

Substituting this back into the first result:

$$\begin{aligned} \int e^x \sin x \, dx &= e^x \sin x - \left(e^x \cos x - \int e^x (-\sin x) \, dx \right) \\ &= e^x \sin x - e^x \cos x - \int e^x \sin x \, dx \end{aligned}$$

Let $I = \int e^x \sin x \, dx$. Then:

$$\begin{aligned} I &= e^x (\sin x - \cos x) - I \\ 2I &= e^x (\sin x - \cos x) \\ I &= \frac{1}{2} e^x (\sin x - \cos x) \end{aligned}$$

Evaluating from 0 to π :

$$\begin{aligned} \int_0^\pi e^x \sin x \, dx &= \left[\frac{1}{2} e^x (\sin x - \cos x) \right]_0^\pi \\ &= \frac{1}{2} (e^\pi (\sin \pi - \cos \pi) - e^0 (\sin 0 - \cos 0)) \\ &= \frac{1}{2} (e^\pi (0 - (-1)) - 1(0 - 1)) \\ &= \frac{1}{2} (e^\pi + 1) \end{aligned}$$

Question 5:

Given $I_n = \int_0^1 x(1 - x^3)^n \, dx$.

We use integration by parts. Let $u = (1 - x^3)^n$ and $dv = x \, dx$.

$$\begin{aligned} du &= n(1 - x^3)^{n-1}(-3x^2) \, dx \\ dv = x \, dx &\implies v = \frac{1}{2} x^2 \end{aligned}$$

Applying integration by parts:

$$\begin{aligned}
I_n &= \left[\frac{1}{2} x^2 (1 - x^3)^n \right]_0^1 - \int_0^1 \frac{1}{2} x^2 \cdot n(1 - x^3)^{n-1} (-3x^2) dx \\
&= 0 - 0 - \int_0^1 \left(-\frac{3n}{2} x^4 (1 - x^3)^{n-1} \right) dx \\
&= \frac{3n}{2} \int_0^1 x^4 (1 - x^3)^{n-1} dx
\end{aligned}$$

To relate this to I_{n-1} , we rewrite x^4 as $x(x^3)$ and use the identity $x^3 = 1 - (1 - x^3)$:

$$\begin{aligned}
I_n &= \frac{3n}{2} \int_0^1 x(x^3)(1 - x^3)^{n-1} dx \\
&= \frac{3n}{2} \int_0^1 x(1 - (1 - x^3))(1 - x^3)^{n-1} dx \\
&= \frac{3n}{2} \left(\int_0^1 x(1 - x^3)^{n-1} dx - \int_0^1 x(1 - x^3)^n dx \right) \\
&= \frac{3n}{2} (I_{n-1} - I_n)
\end{aligned}$$

Solving for I_n :

$$\begin{aligned}
I_n &= \frac{3n}{2} I_{n-1} - \frac{3n}{2} I_n \\
I_n \left(1 + \frac{3n}{2} \right) &= \frac{3n}{2} I_{n-1} \\
I_n \left(\frac{2 + 3n}{2} \right) &= \frac{3n}{2} I_{n-1} \\
I_n &= \frac{3n}{3n + 2} I_{n-1}
\end{aligned}$$

To find I_n in terms of n , we find I_0 : $I_0 = \int_0^1 x(1 - x^3)^0 dx = \int_0^1 x dx = \left[\frac{1}{2} x^2 \right]_0^1 = \frac{1}{2}$
align*

Using the recurrence relation:

$$\begin{aligned}
I_n &= \frac{3n}{3n + 2} \cdot \frac{3(n-1)}{3(n-1) + 2} \cdots \frac{3(1)}{3(1) + 2} I_0 \\
&= \frac{3^n \text{imes } n!}{(3n + 2)(3n - 1) \cdots 5} \cdot \frac{1}{2}
\end{aligned}$$

Skill-building

Question 1:

To evaluate $\int \tan^{-1} x \, dx$, we use integration by parts. Let $u = \tan^{-1} x$ and $dv = 1 \, dx$.

Then, we have:

$$\begin{aligned}
du &= \frac{1}{1 + x^2} dx \\
dv = 1 \, dx &\implies v = x
\end{aligned}$$

Applying the integration by parts formula $\int u dv = uv - \int v du$:

$$\begin{aligned}\int \tan^{-1} x dx &= x \tan^{-1} x - \int \frac{x}{1+x^2} dx \\ &= x \tan^{-1} x - \frac{1}{2} \int \frac{2x}{1+x^2} dx \\ &= x \tan^{-1} x - \frac{1}{2} \ln |1+x^2| + C\end{aligned}$$

Question 2:

We evaluate $\int_0^\pi x^2 \sin\left(\frac{1}{2}x\right) dx$ using integration by parts twice.

Let $u = x^2$ and $dv = \sin\left(\frac{1}{2}x\right) dx$.

$$\begin{aligned}du &= 2x dx \\ dv = \sin\left(\frac{1}{2}x\right) dx &\implies v = -2 \cos\left(\frac{1}{2}x\right)\end{aligned}$$

Applying integration by parts:

$$\begin{aligned}\int x^2 \sin\left(\frac{1}{2}x\right) dx &= -2x^2 \cos\left(\frac{1}{2}x\right) - \int (-2 \cos\left(\frac{1}{2}x\right))(2x) dx \\ &= -2x^2 \cos\left(\frac{1}{2}x\right) + 4 \int x \cos\left(\frac{1}{2}x\right) dx\end{aligned}$$

Now, we apply integration by parts to $\int x \cos\left(\frac{1}{2}x\right) dx$. Let $u = x$ and $dv = \cos\left(\frac{1}{2}x\right) dx$.

$$\begin{aligned}du &= dx \\ dv = \cos\left(\frac{1}{2}x\right) dx &\implies v = 2 \sin\left(\frac{1}{2}x\right)\end{aligned}$$

Substituting this back:

$$\begin{aligned}\int x^2 \sin\left(\frac{1}{2}x\right) dx &= -2x^2 \cos\left(\frac{1}{2}x\right) + 4 \left(2x \sin\left(\frac{1}{2}x\right) - \int 2 \sin\left(\frac{1}{2}x\right) dx \right) \\ &= -2x^2 \cos\left(\frac{1}{2}x\right) + 8x \sin\left(\frac{1}{2}x\right) - 8 \int \sin\left(\frac{1}{2}x\right) dx \\ &= -2x^2 \cos\left(\frac{1}{2}x\right) + 8x \sin\left(\frac{1}{2}x\right) + 16 \cos\left(\frac{1}{2}x\right)\end{aligned}$$

Evaluating from 0 to π :

$$\begin{aligned}\int_0^\pi x^2 \sin\left(\frac{1}{2}x\right) dx &= \left[-2x^2 \cos\left(\frac{1}{2}x\right) + 8x \sin\left(\frac{1}{2}x\right) + 16 \cos\left(\frac{1}{2}x\right) \right]_0^\pi \\ &= \left(-2\pi^2 \cos \frac{\pi}{2} + 8\pi \sin \frac{\pi}{2} + 16 \cos \frac{\pi}{2} \right) - \left(-2(0)^2 \cos 0 + 8(0) \sin 0 + 16 \cos 0 \right) \\ &= (0 + 8\pi(1) + 0) - (0 + 0 + 16(1)) \\ &= 8\pi - 16\end{aligned}$$

Question 3:

Let $I_n = \int \tan^n x dx$. We can rewrite the integrand as $\tan^{n-2} x \cdot \tan^2 x$.

Using the identity $\tan^2 x = \sec^2 x - 1$:

$$\begin{aligned}
I_n &= \int \tan^{n-2} x (\sec^2 x - 1) dx \\
&= \int \tan^{n-2} x \sec^2 x dx - \int \tan^{n-2} x dx \\
&= \int \tan^{n-2} x \sec^2 x dx - I_{n-2}
\end{aligned}$$

For the first integral, let $u = \tan x$, then $du = \sec^2 x dx$:

$$\begin{aligned}
\int \tan^{n-2} x \sec^2 x dx &= \int u^{n-2} du \\
&= \frac{u^{n-1}}{n-1} + C \\
&= \frac{1}{n-1} \tan^{n-1} x + C
\end{aligned}$$

Substituting this back into the expression for I_n :

$$I_n = \frac{1}{n-1} \tan^{n-1} x - I_{n-2}$$

Question 4:

Given $I_n = \int_0^{\frac{\pi}{2}} x^n \sin x dx$.

(i) We use integration by parts. Let $u = x^n$ and $dv = \sin x dx$.

$$\begin{aligned}
du &= nx^{n-1} dx \\
dv = \sin x dx &\implies v = -\cos x
\end{aligned}$$

Applying integration by parts:

$$\begin{aligned}
I_n &= [-x^n \cos x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (-\cos x)(nx^{n-1}) dx \\
&= (0 - 0) + n \int_0^{\frac{\pi}{2}} x^{n-1} \cos x dx
\end{aligned}$$

Now, we apply integration by parts again to $\int_0^{\frac{\pi}{2}} x^{n-1} \cos x dx$. Let $u = x^{n-1}$ and $dv = \cos x dx$.

$$\begin{aligned}
du &= (n-1)x^{n-2} dx \\
dv = \cos x dx &\implies v = \sin x
\end{aligned}$$

Substituting this back:

$$\begin{aligned}
I_n &= n \left([x^{n-1} \sin x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \sin x (n-1)x^{n-2} dx \right) \\
&= n \left(\left(\frac{\pi}{2} \right)^{n-1} \sin 1 - 0 - (n-1) \int_0^{\frac{\pi}{2}} x^{n-2} \sin x dx \right) \\
&= n \left(\left(\frac{\pi}{2} \right)^{n-1} - n(n-1)I_{n-2} \right)
\end{aligned}$$

(ii) To evaluate I_4 , we use the recurrence relation $I_n = n\left(\frac{\pi}{2}\right)^{n-1} - n(n-1)I_{n-2}$:
For $n = 2$:

$$\begin{aligned}
I_2 &= 2 \left(\frac{\pi}{2} \right)^{2-1} - 2(2-1)I_0 \\
&= 2 \left(\frac{\pi}{2} \right) - 2I_0 = \pi - 2I_0
\end{aligned}$$

We find I_0 :

$$I_0 = \int_0^{\frac{\pi}{2}} x^0 \sin x dx = \int_0^{\frac{\pi}{2}} \sin x dx = [-\cos x]_0^{\frac{\pi}{2}} = 0 - (-1) = 1$$

So, $I_2 = \pi - 2(1) = \pi - 2$.

For $n = 4$:

$$\begin{aligned}
I_4 &= 4 \left(\frac{\pi}{2} \right)^{4-1} - 4(4-1)I_2 \\
&= 4 \left(\frac{\pi}{2} \right)^3 - 12(\pi - 2) \\
&= 4 \left(\frac{\pi^3}{8} \right) - 12\pi + 24 \\
&= \frac{\pi^3}{2} - 12\pi + 24
\end{aligned}$$

Easier Exam Questions

Question 1:

To find $\int x e^x dx$, we use integration by parts. Let $u = x$ and $dv = e^x dx$.

Then, we have:

$$\begin{aligned}
du &= dx \\
dv = e^x dx &\implies v = e^x
\end{aligned}$$

Applying the integration by parts formula $\int u dv = uv - \int v du$:

$$\begin{aligned}
\int x e^x dx &= x e^x - \int e^x dx \\
&= x e^x - e^x + C \\
&= e^x(x - 1) + C
\end{aligned}$$

Question 2:

To evaluate $\int_0^2 x^2 \ln x dx$, we use integration by parts. Let $u = \ln x$ and $dv = x^2 dx$.

Then, we have:

$$\begin{aligned}
du &= \frac{1}{x} dx \\
dv = x^2 dx &\implies v = \frac{1}{3} x^3
\end{aligned}$$

Applying the integration by parts formula $\int u dv = uv - \int v du$:

$$\begin{aligned}
\int x^2 \ln x \, dx &= \frac{1}{3}x^3 \ln x - \int \frac{1}{3}x^3 \left(\frac{1}{x}\right) dx \\
&= \frac{1}{3}x^3 \ln x - \frac{1}{3} \int x^2 \, dx \\
&= \frac{1}{3}x^3 \ln x - \frac{1}{9}x^3 + C
\end{aligned}$$

Evaluating from 0 to 2:

$$\begin{aligned}
\int_0^2 x^2 \ln x \, dx &= \left[\frac{1}{3}x^3 \ln x - \frac{1}{9}x^3 \right]_0^2 \\
&= \left(\frac{1}{3}(2)^3 \ln 2 - \frac{1}{9}(2)^3 \right) - \lim_{x \rightarrow 0^+} \left(\frac{1}{3}x^3 \ln x - \frac{1}{9}x^3 \right) \\
&= \left(\frac{8}{3} \ln 2 - \frac{8}{9} \right) - 0 \\
&= \frac{8}{3} \ln 2 - \frac{8}{9}
\end{aligned}$$

Question 3:

Given $I_N = \int_0^{\frac{\pi}{2}} \sin^N x \, dx$.

(i) We use integration by parts. Let $u = \sin^{N-1} x$ and $dv = \sin x \, dx$.

$$\begin{aligned}
du &= (N-1) \sin^{N-2} x \cos x \, dx \\
dv = \sin x \, dx &\implies v = -\cos x
\end{aligned}$$

Applying integration by parts:

$$\begin{aligned}
I_N &= \left[-\sin^{N-1} x \cos x \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (-\cos x)(N-1) \sin^{N-2} x \cos x \, dx \\
&= (0 - 0) + (N-1) \int_0^{\frac{\pi}{2}} \sin^{N-2} x \cos^2 x \, dx \\
&= (N-1) \int_0^{\frac{\pi}{2}} \sin^{N-2} x (1 - \sin^2 x) \, dx \\
&= (N-1) \left(\int_0^{\frac{\pi}{2}} \sin^{N-2} x \, dx - \int_0^{\frac{\pi}{2}} \sin^N x \, dx \right) \\
&= (N-1)(I_{N-2} - I_N)
\end{aligned}$$

Solving for I_N :

$$\begin{aligned}
I_N &= (N-1)I_{N-2} - (N-1)I_N \\
I_N + (N-1)I_N &= (N-1)I_{N-2} \\
NI_N &= (N-1)I_{N-2} \\
I_N &= \frac{N-1}{N} I_{N-2}
\end{aligned}$$

(ii) To find I_6 , we use the recurrence relation $I_N = \frac{N-1}{N} I_{N-2}$:

$$\begin{aligned}
I_6 &= \frac{5}{6} I_4 \\
&= \frac{5}{6} \left(\frac{3}{4} I_{2ight} \right) \\
&= \frac{5}{6} \cdot \frac{3}{4} \left(\frac{1}{2} I_{0ight} \right) \\
&= \frac{5 \cdot 3 \cdot 1}{6 \cdot 4 \cdot 2} I_0
\end{aligned}$$

We find I_0 :

$$I_0 = \int_0^{\frac{\pi}{2}} \sin^0 x \, dx = \int_0^{\frac{\pi}{2}} 1 \, dx = [x]_0^{\frac{\pi}{2}} = \frac{\pi}{2}$$

Substituting I_0 :

$$I_6 = \frac{15}{48} \cdot \frac{\pi}{2} = \frac{5}{16} \cdot \frac{\pi}{2} = \frac{5\pi}{32}$$

Question 4:

Given $y = \ln(x)$, $x \in [1, e]$.

(i) To find $\int_1^e [\ln(x)]^2 \, dx$, we use integration by parts. Let $u = [\ln(x)]^2$ and $dv = 1 \, dx$.

$$du = 2 \ln(x) \cdot \frac{1}{x} dx$$

$$dv = 1 \, dx \implies v = x$$

Applying integration by parts:

$$\begin{aligned}
\int [\ln(x)]^2 \, dx &= x[\ln(x)]^2 - \int x \left(\frac{2 \ln x}{x} \right) dx \\
&= x[\ln(x)]^2 - 2 \int \ln x \, dx
\end{aligned}$$

Using the result $\int \ln x \, dx = x \ln x - x$:

$$\begin{aligned}
\int [\ln(x)]^2 \, dx &= x[\ln(x)]^2 - 2(x \ln x - x) \\
&= x[\ln(x)]^2 - 2x \ln x + 2x
\end{aligned}$$

Evaluating from 1 to e :

$$\begin{aligned}
\int_1^e [\ln(x)]^2 \, dx &= [x[\ln(x)]^2 - 2x \ln x + 2x]_1^e \\
&= (e[\ln e]^2 - 2e \ln e + 2e) - (1[\ln 1]^2 - 2(1) \ln 1 + 2(1)) \\
&= (e(1)^2 - 2e(1) + 2e) - (0 - 0 + 2) \\
&= e - 2
\end{aligned}$$

(ii) The volume of the solid generated by rotating $y = \ln(x)$ about the x -axis is given by:

$$V = \pi \int_1^e [\ln(x)]^2 dx$$

Using the result from part (i):

$$V = \pi(e - 2)$$

Question 5:

Given $I_n = \int_{-3}^0 x^n \sqrt{x+3} dx$.

(i) We use integration by parts. Let $u = x^n$ and $dv = \sqrt{x+3} dx$.

$$du = nx^{n-1} dx$$

$$dv = (x+3)^{1/2} dx \implies v = \frac{2}{3}(x+3)^{3/2}$$

Applying integration by parts:

$$\begin{aligned} I_n &= \left[x^n \cdot \frac{2}{3}(x+3)^{3/2} \right]_{-3}^0 - \int_{-3}^0 \frac{2}{3}(x+3)^{3/2} (nx^{n-1}) dx \\ &= (0 - 0) - \frac{2n}{3} \int_{-3}^0 x^{n-1} (x+3)^{3/2} dx \\ &= -\frac{2n}{3} \int_{-3}^0 x^{n-1} (x+3)(x+3)^{1/2} dx \\ &= -\frac{2n}{3} \int_{-3}^0 (x^n + 3x^{n-1})(x+3)^{1/2} dx \\ &= -\frac{2n}{3} \left(\int_{-3}^0 x^n (x+3)^{1/2} dx + 3 \int_{-3}^0 x^{n-1} (x+3)^{1/2} dx \right) \\ &= -\frac{2n}{3} (I_n + 3I_{n-1}) \end{aligned}$$

Solving for I_n :

$$\begin{aligned} I_n &= -\frac{2n}{3} I_n - 2n I_{n-1} \\ I_n + \frac{2n}{3} I_n &= -2n I_{n-1} \\ I_n \left(1 + \frac{2n}{3} \right) &= -2n I_{n-1} \\ I_n \left(\frac{3+2n}{3} \right) &= -2n I_{n-1} \\ I_n &= -\frac{6n}{2n+3} I_{n-1} \end{aligned}$$

(ii) To find I_3 , we use the recurrence relation $I_n = -\frac{6n}{2n+3} I_{n-1}$:

$$\begin{aligned} I_3 &= -\frac{6(3)}{2(3)+3} I_2 = -\frac{18}{9} I_2 = -2I_2 \\ I_2 &= -\frac{6(2)}{2(2)+3} I_1 = -\frac{12}{7} I_1 \\ I_1 &= -\frac{6(1)}{2(1)+3} I_0 = -\frac{6}{5} I_0 \end{aligned}$$

We find I_0 :

$$I_0 = \int_{-3}^0 (x+3)^{1/2} dx = \left[\frac{2}{3} (x+3)^{3/2} \right]_{-3}^0 = \frac{2}{3} (3)^{3/2} - 0 = \frac{2}{3} \text{ times } 3\sqrt{3} = 2\sqrt{3}$$

Substituting back:

$$\begin{aligned} I_1 &= -\frac{6}{5} (2\sqrt{3}) = -\frac{12\sqrt{3}}{5} \\ I_2 &= -\frac{12}{7} \left(-\frac{12\sqrt{3}}{5} \right) = \frac{144\sqrt{3}}{35} \\ I_3 &= -2 \left(\frac{144\sqrt{3}}{35} \right) = -\frac{288\sqrt{3}}{35} \end{aligned}$$

Harder Exam Questions

Question 1:

Given $I = \int_2^3 \tan^{-1} \left(\frac{1}{x^2 - 5x + 7} \right) dx$.

(i) We use the identity $\tan^{-1} A - \tan^{-1} B = \tan^{-1} \left(\frac{A-B}{1+AB} \right)$.

Let $A = x - 2$ and $B = x - 3$.

$$\begin{aligned} \tan^{-1}(x-2) - \tan^{-1}(x-3) &= \tan^{-1} \left(\frac{(x-2) - (x-3)}{1 + (x-2)(x-3)} \right) \\ &= \tan^{-1} \left(\frac{1}{1 + (x^2 - 5x + 6)} \right) \\ &= \tan^{-1} \left(\frac{1}{x^2 - 5x + 7} \right) \end{aligned}$$

(ii) Now we evaluate I :

$$\begin{aligned} I &= \int_2^3 (\tan^{-1}(x-2) - \tan^{-1}(x-3)) dx \\ &= \int_2^3 \tan^{-1}(x-2) dx - \int_2^3 \tan^{-1}(x-3) dx \end{aligned}$$

Let $u = x - 2$ in the first integral and $v = x - 3$ in the second integral.

$$\begin{aligned} \int \tan^{-1}(x-2) dx &= (x-2) \tan^{-1}(x-2) - \int (x-2) \frac{1}{1 + (x-2)^2} dx \\ &= (x-2) \tan^{-1}(x-2) - \frac{1}{2} \ln(1 + (x-2)^2) \end{aligned}$$

Similarly,

$$\int \tan^{-1}(x-3) dx = (x-3) \tan^{-1}(x-3) - \frac{1}{2} \ln(1 + (x-3)^2)$$

Evaluating I :

$$\begin{aligned}
I &= \left[(x-2) \tan^{-1}(x-2) - \frac{1}{2} \ln(1+(x-2)^2) - (x-3) \tan^{-1}(x-3) + \frac{1}{2} \ln(1+(x-3)^2) \right]_2^3 \\
&= \left(\tan^{-1}(1) - \frac{1}{2} \ln(2) - (0) \tan^{-1}(0) + \frac{1}{2} \ln(1) \right) - \left((0) \tan^{-1}(0) - \frac{1}{2} \ln(1) - (-1) \tan^{-1}(-1) + \frac{1}{2} \ln(2) \right) \\
&= \left(\frac{\pi}{4} - \frac{1}{2} \ln 2 + 0 + 0 \right) - \left(0 - 0 - (-1) \left(-\frac{\pi}{4} \right) + \frac{1}{2} \ln 2 \right) \\
&= \frac{\pi}{4} - \frac{1}{2} \ln 2 - \left(-\frac{\pi}{4} + \frac{1}{2} \ln 2 \right) \\
&= \frac{\pi}{2} - \ln 2
\end{aligned}$$

Question 2:

The argument attempts to prove $0 = 1$ by evaluating $\int \frac{1}{x} dx$ using integration by parts.

Let's examine the steps:

$$\begin{aligned}
\int \frac{1}{x} dx &= \int \frac{1}{x} \times 1 dx \\
&= \frac{1}{x} \times x - \int \left(-\frac{1}{x^2} \right) x dx \\
&= 1 + \int \frac{1}{x} dx
\end{aligned}$$

The error lies in the treatment of the constant of integration. When we write $\int \frac{1}{x} dx = 1 + \int \frac{1}{x} dx$, we are dealing with indefinite integrals. An indefinite integral represents a family of functions. The equality $\int \frac{1}{x} dx = f(x) + C_1$ and $\int \frac{1}{x} dx = f(x) + C_2$ is valid if C_1 and C_2 only differ by a constant.

Specifically, the integration by parts formula $\int u dv = uv - \int v du$ includes an implicit constant of integration. In the step $\int \frac{1}{x} dx = 1 + \int \frac{1}{x} dx$, the two integrals on both sides are the same function but shifted by a constant. Subtracting the integral from both sides is only valid for specific values (definite integrals) or if we account for the constant C .

If we include the constants:

$$\begin{aligned}
\ln |x| + C_1 &= 1 + \ln |x| + C_2 \\
C_1 &= 1 + C_2
\end{aligned}$$

This simply means the constants of integration differ by 1, which is not a contradiction. Therefore, the argument that $0 = 1$ is invalid because it ignores the constants of integration.

Question 3:

To find $\int \sec^3 x dx$, we use integration by parts. Let $u = \sec x$ and $dv = \sec^2 x dx$.

Then, we have:

$$\begin{aligned}
du &= \sec x \tan x dx \\
dv = \sec^2 x dx &\implies v = \tan x
\end{aligned}$$

Applying integration by parts:

$$\begin{aligned}
\int \sec^3 x dx &= \sec x \tan x - \int \tan x (\sec x \tan x) dx \\
&= \sec x \tan x - \int \sec x \tan^2 x dx
\end{aligned}$$

Using the identity $\tan^2 x = \sec^2 x - 1$:

$$\begin{aligned}\int \sec^3 x \, dx &= \sec x \tan x - \int \sec x (\sec^2 x - 1) \, dx \\ &= \sec x \tan x - \int \sec^3 x \, dx + \int \sec x \, dx\end{aligned}$$

Let $I = \int \sec^3 x \, dx$. Then:

$$\begin{aligned}I &= \sec x \tan x - I + \int \sec x \, dx \\ 2I &= \sec x \tan x + \int \sec x \, dx\end{aligned}$$

Using the standard integral $\int \sec x \, dx = \ln |\sec x + \tan x| + C$:

$$\begin{aligned}2I &= \sec x \tan x + \ln |\sec x + \tan x| \\ I &= \frac{1}{2}(\sec x \tan x + \ln |\sec x + \tan x|) + C\end{aligned}$$

Question 4:

Given $I_{3n+2} = \int_0^1 x^{3n+2} e^{x^3} \, dx$.

(i) We use integration by parts. Let $u = e^{x^3}$ and $dv = x^{3n+2} \, dx$.

$$\begin{aligned}du &= 3x^2 e^{x^3} \, dx \\ dv = x^{3n+2} \, dx &\implies v = \frac{1}{3n+3} x^{3n+3}\end{aligned}$$

Applying integration by parts:

$$\begin{aligned}I_{3n+2} &= \left[e^{x^3} \frac{x^{3n+3}}{3n+3} \right]_0^1 - \int_0^1 \frac{x^{3n+3}}{3n+3} (3x^2 e^{x^3}) \, dx \\ &= \frac{e}{3n+3} - \frac{3}{3n+3} \int_0^1 x^{3n+5} e^{x^3} \, dx\end{aligned}$$

Wait, let's try a different choice of u and dv to get the desired recurrence relation. Let $u = x^{3n-1} e^{x^3}$ and $dv = x^3 \, dx$. No, that's not it. Let's try $u =$

$x^{3n-1} e^{x^3}$ and $dv = 3x^2 e^{x^3} \, dx$. No.

Let's try $u = e^{x^3}$ and $dv = x^{3n+2} \, dx$. The result was $I_{3n+2} = \frac{e}{3n+3} - \frac{1}{n+1} I_{3n+5}$. This is not the requested relation. Let's re-examine the question. The relation is $I_{3n+2} = \frac{e}{3} - n I_{3n-1}$.

Let's try integration by parts on $I_{3n-1} = \int_0^1 x^{3n-1} e^{x^3} \, dx$. Let $u = x^{3n-1}$ and $dv = 3x^2 e^{x^3} \, dx$. Then $v = e^{x^3}$.

$$\begin{aligned}I_{3n-1} &= \int_0^1 x^{3n-1} e^{x^3} \, dx \\ &= \int_0^1\end{aligned}$$

$$x^{3n-3} (3x^2 e^{x^3}) \, dx$$

Let $u = \frac{1}{3} x^{3n-3}$ and $dv = 3x^2 e^{x^3} \, dx$.

$$\begin{aligned}du &= (n-1) x^{3n-4} \, dx \\ dv = 3x^2 e^{x^3} \, dx &\implies v = e^{x^3}\end{aligned}$$

This is also not leading to the relation. Let's try $u = x^{3n+2}$ and $dv = e^{x^3} dx$. No.

Let's try $u = e^{x^3}$ and $dv = x^{3n+2} dx$ again, but maybe the relation is $I_{3n+2} = \frac{e}{3} - nI_{3n-1}$? Let's check $n = 0$. $I_2 = \int_0^1 x^2 e^{x^3} dx = [\frac{1}{3} e^{x^3}]_0^1 = \frac{e-1}{3}$. The formula gives $I_2 = \frac{e}{3} - 0(I_{-1}) = \frac{e}{3}$. Close, but not quite. There might be a typo in the question's recurrence relation. Let's assume the relation is $I_{3n+2} = \frac{e}{3} - nI_{3n-1}$ and proceed to part (ii).

(ii) To find I_8 , we use $n = 2$ in $I_{3n+2} = \frac{e}{3} - nI_{3n-1}$.

$$I_8 = I_{3(2)+2} = \frac{e}{3} - 2I_{3(2)-1} = \frac{e}{3} - 2I_5$$

To find I_5 , we use $n = 1$ in the relation:

$$I_5 = I_{3(1)+2} = \frac{e}{3} - 1I_{3(1)-1} = \frac{e}{3} - I_2$$

We know $I_2 = \int_0^1 x^2 e^{x^3} dx = \frac{e-1}{3}$.

$$I_5 = \frac{e}{3} - \frac{e-1}{3} = \frac{1}{3}$$

Substituting I_5 into I_8 :

$$I_8 = \frac{e}{3} - 2\left(\frac{1}{3}\right) = \frac{e-2}{3}$$

Question 5:

Given $I_n = \int_0^{\frac{\pi}{3}} \tan^{n+1} x \sec x dx$. Given $\int \tan x \sec x dx = \sec x$.

(i) We use integration by parts. Let $u = \tan^n x$ and $dv = \tan x \sec x dx$.

$$du = n \tan^{n-1} x \sec^2 x dx$$

$$dv = \tan x \sec x dx \implies v = \sec x$$

Applying integration by parts:

$$\begin{aligned} I_n &= [\tan^n x \sec x]_0^{\frac{\pi}{3}} - \int_0^{\frac{\pi}{3}} \sec x (n \tan^{n-1} x \sec^2 x) dx \\ &= \left(\tan^n \frac{\pi}{3} \sec \frac{\pi}{3}\right) - 0 - n \int_0^{\frac{\pi}{3}} \tan^{n-1} x \sec^3 x dx \end{aligned}$$

Using $\tan \frac{\pi}{3} = \sqrt{3}$ and $\sec \frac{\pi}{3} = 2$:

$$\begin{aligned} I_n &= 2(\sqrt{3})^n - n \int_0^{\frac{\pi}{3}} \tan^{n-1} x (\sec^2 x + 1) \sec x dx \\ &= 2 \cdot 3^{n/2} - n \left(\int_0^{\frac{\pi}{3}} \tan^{n+1} x \sec x dx + \int_0^{\frac{\pi}{3}} \tan^{n-1} x \sec x dx \right) \\ &= 2 \cdot 3^{n/2} - n(I_n + I_{n-2}) \end{aligned}$$

Solving for I_n :

$$\begin{aligned} I_n + nI_n &= 2 \cdot 3^{n/2} - nI_{n-2} \\ (n+1)I_n &= 2 \cdot 3^{n/2} - nI_{n-2} \\ I_n &= \frac{2 \cdot 3^{n/2}}{n+1} - \frac{n}{n+1} I_{n-2} \end{aligned}$$

(ii) To find I_2 , we use the recurrence relation $I_n = \frac{2 \cdot 3^{n/2}}{n+1} - \frac{n}{n+1} I_{n-2}$:
For $n = 2$:

$$\begin{aligned} I_2 &= \frac{2 \cdot 3^{2/2}}{2+1} - \frac{2}{2+1} I_0 \\ &= \frac{2 \cdot 3}{3} - \frac{2}{3} I_0 = 2 - \frac{2}{3} I_0 \end{aligned}$$

We find I_0 :

$$I_0 = \int_0^{\pi/3} \tan x \sec x \, dx = [\sec x]_0^{\pi/3} = 2 - 1 = 1$$

Substituting I_0 :

$$I_2 = 2 - \frac{2}{3}(1) = \frac{4}{3}$$

Question 6:

Given $I_n = \int_0^{\pi/4} \tan^n x \, dx$.

(i) We use the identity $\tan^2 x = \sec^2 x - 1$:

$$\begin{aligned} I_n &= \int_0^{\pi/4} \tan^{n-2} x (\sec^2 x - 1) \, dx \\ &= \int_0^{\pi/4} \tan^{n-2} x \sec^2 x \, dx - \int_0^{\pi/4} \tan^{n-2} x \, dx \\ &= \int_0^{\pi/4} \tan^{n-2} x \sec^2 x \, dx - I_{n-2} \end{aligned}$$

For the first integral, let $u = \tan x$, then $du = \sec^2 x \, dx$:

$$\int_0^{\pi/4} \tan^{n-2} x \sec^2 x \, dx = \int_0^1 u^{n-2} \, du = \left[\frac{u^{n-1}}{n-1} \right]_0^1 = \frac{1}{n-1}$$

Substituting this back:

$$I_n = \frac{1}{n-1} - I_{n-2} \implies I_n + I_{n-2} = \frac{1}{n-1}$$

(ii) To evaluate I_7 , we use the recurrence relation $I_n + I_{n-2} = \frac{1}{n-1}$:

$$\begin{aligned} I_7 &= \frac{1}{7-1} - I_5 = \frac{1}{6} - I_5 \\ I_5 &= \frac{1}{5-1} - I_3 = \frac{1}{4} - I_3 \\ I_3 &= \frac{1}{3-1} - I_1 = \frac{1}{2} - I_1 \end{aligned}$$

We find I_1 :

$$I_1 = \int_0^{\pi/4} \tan x \, dx = [-\ln |\cos x|]_0^{\pi/4} = -\ln \frac{1}{\sqrt{2}} - (-\ln 1) = \ln \sqrt{2} = \frac{1}{2} \ln 2$$

Substituting back:

$$\begin{aligned}I_3 &= \frac{1}{2} - \frac{1}{2} \ln 2 \\I_5 &= \frac{1}{4} - \left(\frac{1}{2} - \frac{1}{2} \ln 2\right) = \frac{1}{2} \ln 2 - \frac{1}{4} \\I_7 &= \frac{1}{6} - \left(\frac{1}{2} \ln 2 - \frac{1}{4}\right) = \frac{1}{6} + \frac{1}{4} - \frac{1}{2} \ln 2 = \frac{5}{12} - \frac{1}{2} \ln 2\end{aligned}$$